

Customer Shopping Behavior Analysis

End-to-end retail data analytics — Python · SQL · PostgreSQL · Power BI

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Project Objectives

01

Clean & Preprocess

Python & Pandas

02

Store Data

PostgreSQL

03

Query Insights

SQL analysis

04

Visualize

Power BI dashboard

05

Identify Trends

Categories, gender, seasonality

Tools & Technologies

Python

Pandas, NumPy

Jupyter

Notebook

PostgreSQL

Database

SQL

Queries

Power BI

Dashboard

CSV

Dataset

Dataset at a Glance

3,900

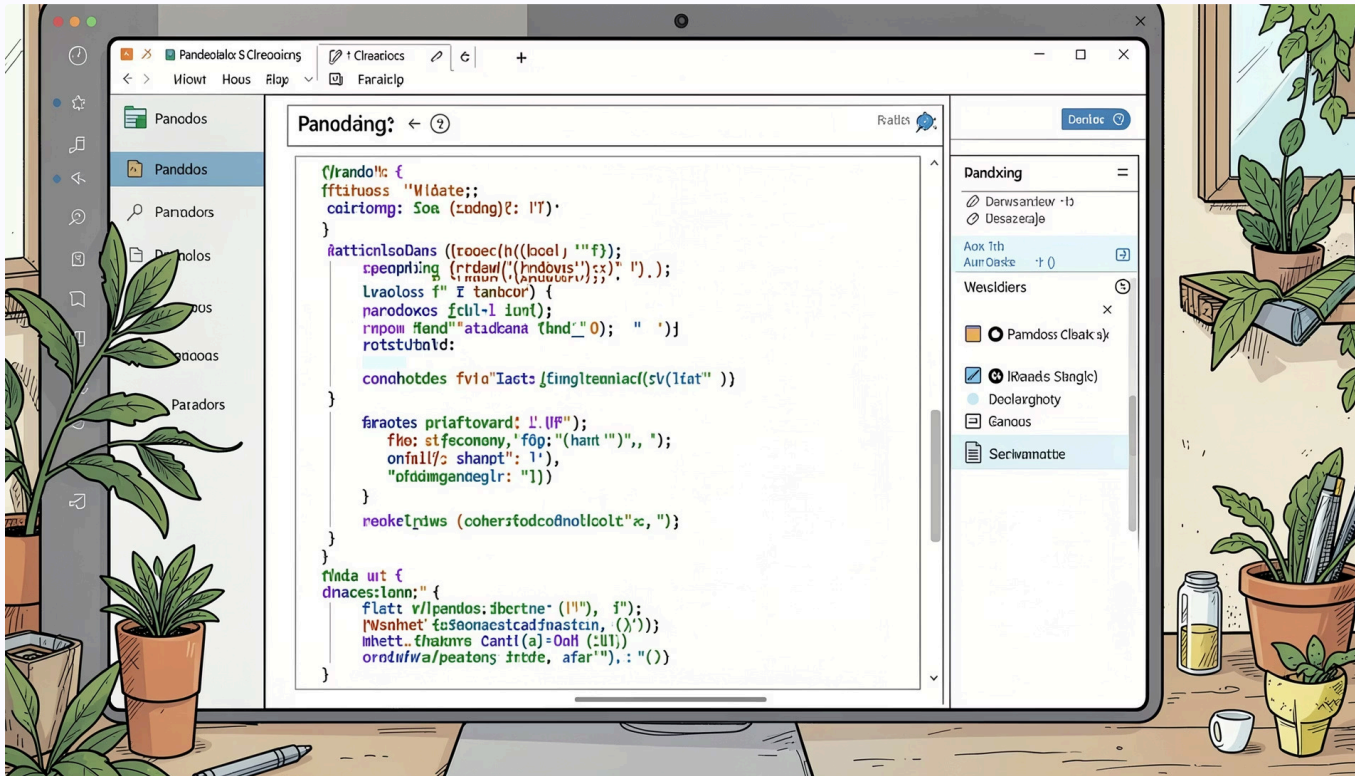
Total Records

18

Columns

Columns spanned customer demographics, product categories, purchase amounts, payment methods, review ratings, and subscription status.

Data Cleaning & Preprocessing



→ Import dataset

Pandas

→ Handle missing values

Null entries

→ Create analytical columns

Categorization

→ Correct datatypes

Formatting

→ Prepare for integration

SQL & Power BI

Python Data Cleaning

Step-by-step cleaning workflow in Jupyter Notebook using Pandas.

[illegible]

```
>>> !python_jupyter notebook > # customer behaviour graph > import pandas as pd
In [All Outputs] | [Outline] --

# median of each category
g = g.fillna(g.groupby('category')['Review Rating'].transform('median'))

# replace '-' with ''
cat_list_end = purchase_amount[]

new_purchase = 'category',
n = 'color', 'venue',
m = 'collaborative_type',
d = 'previous_purchases',
chairs },

including four category - young , adult , middle-age , senior using 'Q Q00'

age','senior']
```

```

> Python_Jupyter_notebook > customer_behaviour_gorb > import pandas as pd
See All Outputs | Outline
group

size_keys"  # .map converts text into numerical

frequency_of_purchases'].map(frequency_mapping)

x_of_purchases'][:head(10)]

purchases
roughly
roughly
Weekly
Weekly
Monthly
Weekly
Quarterly
Weekly

```

```

net > Python_ayher notebook > customer_behavior.py > pip install pyprojct binary sqllchemy
See All Outputs | Outline
ip, frequency_purchase_days'].

real site-packages is not writable
Error in C:\Users\alishah\AppData\Localing\Packages\PythonSoftwareFoundation.Python.3.9.12\q...
In C:\Users\alishah\AppData\Localing\Packages\PythonSoftwareFoundation.Python.3.9.12\q...
In C:\Users\alishah\AppData\Localing\Packages\PythonSoftwareFoundation.Python.3.9.12\q...
pipenv-6.8.0 In C:\Users\alishah\AppData\Localing\Packages\PythonSoftwareFoundation.Python.3.9.12\q...
is not writable package.
> 26.1.1 -> 26.1.2
> install --upgrade pip

press database

ddag@localhost:5432/pyprojct_customer_behavior"

ml
```

[illegible]

SQL Analysis

Revenue by Gender & Category

Top-Performing Categories

Seasonal Sales Trends

Subscription Behavior

Customer Purchasing Insights



SQL Query Results

SQL Query Editor

```
(SELECT AVG(purchase_amount) FROM customer);
```

Customers into New, Returning, and Loyal based on their total number of previous purchases and total purchase amount of each segment

```
WITH customer_segments AS (
  SELECT customer_id, previous_purchases,
         CASE
           WHEN previous_purchases = 1 THEN 'New'
           WHEN previous_purchases BETWEEN 2 AND 10 THEN 'Returning'
           ELSE 'Loyal'
         END AS segment
  FROM customer
)
SELECT segment, COUNT(*) AS "number_of_customers"
FROM customer_segments
GROUP BY segment;
```

segment	number_of_customers
New	83
Returning	3116
Loyal	701

SQL Query Editor

```
What are the top 3 most purchased products within each category?
```

```
WITH item_rank AS (
  SELECT category, item_purchased, total_orders
  FROM purchases
  ORDER BY category, item_purchased
)
SELECT item_rank, category, item_purchased, total_orders
FROM item_rank
WHERE item_rank <= 3;
```

Messages Notifications

category	item_purchased	total_orders
Accessories	Jewelry	171
Accessories	Sunglasses	161
Accessories	Belt	161
Clothing	Blouse	171
Clothing	Pants	171
Clothing	Shirt	169
Footwear	Sandals	160
Footwear	Shoes	180
Footwear	Sneakers	145
Outerwear	Jacket	163
Outerwear	Coat	161

SQL Query Editor

```
5 products have the highest percentage of purchases with discounts applied?
```

```
WITH discount_rate AS (
  SELECT item_purchased,
         SUM(CASE WHEN discount_applied = 'Yes' THEN 1 ELSE 0 END) / COUNT(*) AS discount_rate
  FROM purchases
  GROUP BY item_purchased
)
SELECT item_purchased, discount_rate
FROM discount_rate
ORDER BY discount_rate DESC
LIMIT 5;
```

Messages Notifications

item_purchased	discount_rate
Accessories	50.00
Accessories	49.66
Accessories	49.07
Accessories	48.17
Accessories	47.37

SQL Query Editor

```
Which customers spend more? Compare average spend and total revenue between subscription status.
```

```
WITH customer_stats AS (
  SELECT subscription_status,
         COUNT(*) AS total_customers,
         SUM(purchase_amount) / COUNT(*) AS average_spend,
         SUM(purchase_amount) AS total_revenue
  FROM customer
  GROUP BY subscription_status
)
SELECT subscription_status,
       average_spend, total_revenue
FROM customer_stats
ORDER BY total_revenue DESC;
```

Messages Notifications

subscription_status	total_customers	average_spend	total_revenue
Subscriber	1053	59.49	62645.00
Non-Subscriber	2847	59.87	170436.00

Power BI Dashboard



Dashboard Components

KPI Cards

Seasonal Revenue Charts

Customer Demographics

Payment Methods

Subscription Analysis

Category Performance

Key Insights & Conclusion



Clothing

Highest revenue category



Male Customers

Larger revenue share



Seasonality

Strong seasonal patterns



Subscriptions

Distinct purchasing behavior



Payments

Nearly equal preference

End-to-end workflow: raw data → cleaned dataset → SQL insights → Power BI dashboard. Hands-on experience with Python, PostgreSQL, SQL, and Power BI.